programmable interface device

- A programmable interface device is designed to perform various input/output functions.
- can be set to perform specific functions by writing an instruction in Its internal register (control register.)
- These devices are flexible. versatile. and economical:
- widely used in microprocessor based products.

3.1:Basic Concepts in Programmable Devices:

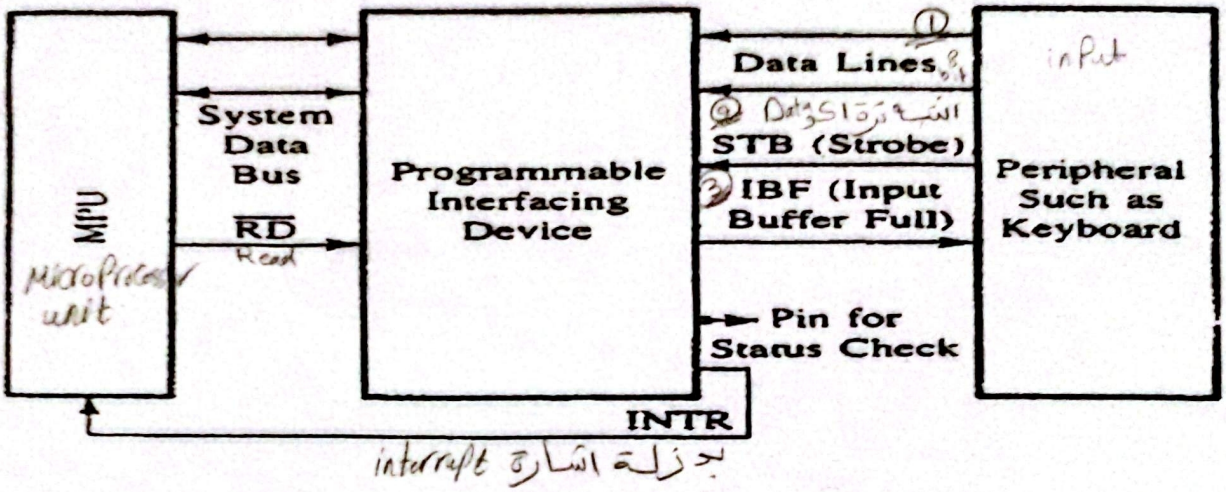
The MPU needs to check a peripheral is ready before it reads from or writes into a device because the execution speed of the microprocessor is much faster than the response of a peripheral such as a printer. To prevent the loss of data or the MPU reading the same data more than once, signals are exchanged between the MPU and a peripheral prior to actual data transfer(called handshake signals), which provided by the interfacing circuitry.

The requirements for a programmable interfacing device:

1. Input and output registers (a group of latches to hold data).
2. Tri-state buffers.
3. Capability for bidirectional data flow.
4. Handshake and interrupt signals.
5. Control logic.
6. Chip select logic.
7. Interrupt control logic.

3.2 Programmable Devices with Handshake Signals

3.2.1 Data Input With Handshake



The steps in data input from a peripheral such as a keyboard are as follows:

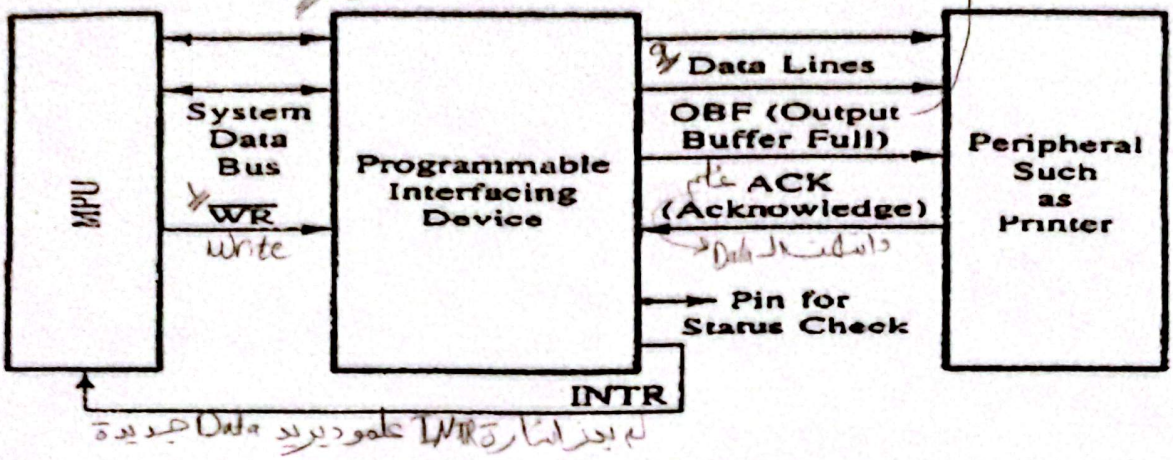
1) A peripheral strobes or places a data byte in the input port and informs the interfacing device by sending handshake signal STB (Strobe).

2) The device informs the peripheral that its input port is full-do not send the next byte until this one has been read. This message is conveyed to the peripheral by sending handshake signal IBF (Input Buffer Full).

3) The MPU keeps checking the status until a byte is available. Or the interfacing device informs the MPU, by sending an interrupt, that it has a byte to be read.

4) The MPU reads the byte by sending control signal $\overline{RD}$

3.2.2 Data Output With Handshake



Fig(3.4): Interfacing Device with Handshake Signals for Data output

1) The MPU writes a byte into the output port of the programmable device by sending control signal $\overline{RW}$
2) The device informs the peripheral, by sending handshake signal OBF (Output Buffer Full), that a byte is on the way.
3) The peripheral acknowledges the byte by sending back the ACK (Acknowledge) signal to the device.
4) The device interrupts the MPU to ask for the next byte, or the MPU finds out that the byte has been acknowledged through the status check.